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Adaptive QKD link establishment for single- and dual-layer LEO quantum satellite networks

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Satellite-based quantum key distribution (QKD), especially relying on low Earth orbit (LEO) satellites, overcomes the geographical limitations of fiber-based QKD and enables global secure communications. However, existing QKD link establishment schemes for LEO quantum satellite networks fail to fully consider practical scenarios (e.g., multi-layer constellations as well as heterogeneous satellite–ground and inter-satellite QKD links) along with realistic constraints (e.g., different illumination conditions and the need to avoid frequent link switching). To address this gap, this work focuses on the adaptive establishment of satellite–ground and inter-satellite QKD links in single- and dual-layer LEO quantum satellite networks. We describe the network structures of single- and dual-layer constellations, as well as formulate the corresponding network model, secret key rate model, and secret key volume model. To realize adaptive QKD link establishment, we develop two integer linear programming (ILP) models and design three heuristic algorithms. The two ILP models are denoted as ILP-G and ILP-R, respectively, where ILP-G aims to maximize the generated key volume, and ILP-R is dedicated to optimizing global key resource balance by prioritizing QKD links with lower remaining key volume. The three heuristic algorithms are designed based on maximum key rate (MaxKR), maximum generated key volume (MaxGKV), and minimum remaining key volume (MinRKV). Simulation results verify the applicability of the proposed approaches across single- and dual-layer LEO quantum satellite networks. Comparative evaluations against two benchmarks, i.e., random fit (RF) and first fit (FF), show that ILP-G and MaxGKV achieve the largest cumulative key volume, while ILP-R and MinRKV outperform the other approaches in key volume distribution, quantum satellite availability, and total number of established QKD links. Specifically, the cumulative key volume of ILP-G is 44.0% and 33.8% higher than that of RF in single- and dual-layer LEO quantum satellite networks, respectively. Furthermore, the dual-layer constellation generates more key resources, with the cumulative key volume reaching approximately five times that of the single-layer constellation. It also exhibits a more balanced global key resource distribution, where key resources are distributed across a broader set of node pairs rather than concentrated on a limited subset of nodes.

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1. INTRODUCTION

With the continuous advancements of quantum computing [1,2], the vulnerability of traditional cryptographic techniques has become increasingly prominent. This poses significant challenges to the long-term security of communication systems relying on the classical cryptography. Quantum key distribution (QKD) [3,4], grounded in the principles of quantum physics, offers a robust solution for future-proof security. By establishing a QKD link between two communicating parties, information-theoretically secure key negotiation can be achieved. In practice, QKD networks are mainly constructed via fiber-based or satellite-based QKD links. Fiber-based QKD has been deployed in multiple metropolitan networks and backbone networks [5]. However, due to the physical constraints of fiber deployment, its application in large-scale, long-distance communication scenarios still faces certain limitations. In contrast, satellite-based QKD can effectively break through geographical barriers and demonstrate substantial application potential for global secure communications [6–8].

Low Earth orbit (LEO) satellites, typically operating at altitudes between 300 and 2000 km, offer advantages such as low cost and relatively short communication distances, making them an effective platform for satellite-based QKD. Several practical experiments relying on LEO quantum satellites (e.g., Micius [9,10] and Jinan-1 [11]) have verified their technical feasibility. Furthermore, to achieve global key distribution, research on constructing LEO constellations for quantum satellite networks has gradually gained momentum [12–15]. In existing studies, not only are single-layer LEO quantum satellite constellations with identical orbits (e.g., the typical polar orbit) considered, but multi-layer LEO quantum satellite constellations with orbits of varying inclinations have also garnered growing attention [16]. Notably, the multi-layer LEO constellation has been widely adopted for classical satellite communications, which has demonstrated improved quality of service (QoS) and seamless coverage [17–19]. Consequently, it is important to conduct in-depth comparative studies on the performance differences between single-layer and multi-layer LEO quantum satellite networks.

In quantum satellite networks, QKD links can be established between two nodes (e.g., a quantum satellite and a ground station, or two quantum satellites), which are mainly categorized into satellite–ground QKD links (SGQLs) and inter-satellite QKD links (ISQLs). Notably, for SGQLs, downlink and uplink show distinct performance characteristics. In the downlink (i.e., from the satellite to the ground), beam wandering induced by atmospheric turbulence occurs near the Earth's surface, where the diffraction-limited beam size is typically much larger than the wandering effect. This leads to reduced beam spreading and higher link efficiency compared to the uplink, with the downlink already being implemented in practical satellite-based QKD systems [9–11]. In contrast to static terrestrial fiber-based networks, the relative positions between quantum satellites and ground stations, or between quantum satellites themselves, undergo real-time dynamic changes, which directly lead to the potential link interruption or reestablishment over time. Thus, determining an appropriate QKD link establishment scheme is crucial for realizing efficient satellite-based QKD. Existing studies have proposed corresponding satellite–ground link establishment schemes for entanglement distribution links [20–24] and entanglement-based QKD links [25]. However, current QKD link establishment schemes still fail to fully consider the practical scenarios and constraints of LEO quantum satellite networks. For example, insufficient attention has been paid to the establishment of ISQLs and the adoption of more practical QKD protocols such as the BB84 protocol [26] used by the Micius and Jinan-1 quantum satellites. Additionally, some realistic constraints have not been fully incorporated, such as the impact of spatial and terrestrial background light on key rates, as well as the negative impact of frequent link switching on the continuity of key negotiation. In this regard, the coordinated establishment of SGQLs and ISQLs based on more realistic constraints remains to be further explored.

This work aims to address the QKD link establishment problem in single- and dual-layer LEO quantum satellite networks, where the establishment of SGQLs and ISQLs is holistically considered in conjunction with various real-world constraints. In particular, a series of adaptive QKD link establishment approaches are proposed. The major contributions of this work are summarized as follows:

1. We describe the network structures for single- and dual-layer LEO quantum satellite networks, where the Walker-star and Walker-delta constellations are combined to form a dual-layer constellation for achieving better complementarity.
2. We formulate the corresponding network model, secret key rate (SKR) model, and secret key volume (SKV) model for SGQLs and ISQLs, where the BB84 protocol is adopted, as well as different illumination conditions and the total transmission efficiency under various influencing factors are incorporated.
3. We develop two integer linear programming (ILP) models (namely, ILP-G and ILP-R) for optimizing QKD link establishment in LEO quantum satellite networks, with objectives of maximizing the generated key volume (GKV) and balancing the remaining key volume (RKV), respectively, both of which are subject to practical constraints such as the limited number of QKD devices at quantum satellites and ground stations.
4. We devise three adaptive QKD link establishment schemes and design their corresponding heuristic algorithms, namely, the maximum key rate-based (MaxKR), the maximum generated key volume-based (MaxGKV), and the minimum remaining key volume-based (MinRKV) QKD link establishment algorithms.
5. We verify the applicability of the presented ILP models and heuristic algorithms for single- and dual-layer LEO quantum satellite networks, as well as compare them against two benchmarks in terms of cumulative key volume, key volume distribution, quantum satellite availability, and the total number of established QKD links.

The rest of this paper is organized as follows. Section 2 briefly reviews the related work. Section 3 describes the network structure as well as formulates the network model, SKR model, and SKV model. Section 4 develops the two ILP models. The three adaptive QKD link establishment schemes and their corresponding heuristic algorithms are presented in Section 5. Section 6 conducts simulations and performs the detailed comparative evaluation. Finally, Section 7 concludes this paper.

2. RELATED WORK

In terms of quantum satellites, LEO satellites are currently the most practical platforms due to their low cost and short distance to the ground, which facilitates relatively efficient quantum signal transmission. The Micius quantum satellite first validated LEO satellite-based QKD [9] and demonstrated satellite-based intercontinental QKD networking [10], while the Jinan-1 quantum satellite was a smaller-sized and lighter-weight LEO microsatellite that realized real-time QKD [11]. Both satellites employed the BB84 protocol [26] proposed by Bennett and Brassard in 1984. In recent years, quantum satellites and satellite-based QKD networks are being developed and planned in many countries [8], most of which also consider LEO satellites as their first choice.

Furthermore, quantum satellites are expected to be interconnected to form quantum satellite networks that support global secure communications. An increasing number of studies have emerged in this field. For the connection of core fiber networks and ground stations [27], as well as various metropolitan networks [28], the configurations of quantum satellites have been optimized, and their efficiencies have been confirmed. In addition, Guo *et al.* [29] fully exploited the wide coverage advantage of geostationary-Earth-orbit (GEO) classical communication satellites to assist in key relaying in LEO quantum satellite networks. He *et al.* [30] proposed a satellite relay tree-based key provisioning approach for securing multicast services in quantum satellite networks. As a fundamental problem, the modeling and design of LEO quantum satellite constellations have been investigated in [12–15], where single-layer constellations are commonly considered and multi-layer constellations are preliminarily explored. These studies have demonstrated the feasibility and superiority of constructing LEO quantum satellite networks to provide QKD capabilities across worldwide regions. Specifically, Li *et al.* [16] experimentally demonstrated that dual-layer LEO quantum satellite networks comprising the medium-inclination orbit (e.g., the inclination ranging from 30° to 60° or 120° to 150°) and sun-synchronous orbit (i.e., the inclination of approximately 90°) can exhibit strong complementarity.

With regard to satellite-based QKD links, they can be categorized into SGQLs and ISQLs. The SGQLs primarily handle key exchange between quantum satellites and ground stations. The feasibility of SGQLs has been experimentally validated using practical LEO quantum satellites and satellite-based QKD networks [10,31]. The ISQLs are mainly used to produce secret keys between two quantum satellites, which can balance key resources and thereby ensure continuous key coverage across the global QKD network. Although ISQLs have not been demonstrated using practical quantum satellites, their feasibility has been demonstrated in long-distance free-space QKD experiments [32]. Furthermore, Grossi *et al.* [15] validated via modeling that inter-planar and intra-planar ISQLs can significantly enhance daily key throughput. In addition, the impact of sunlight on the establishment of QKD links has been studied, and the feasibility of daylight QKD has been verified in numerous free-space QKD experiments [32–36]. However, it remains challenging to implement daylight satellite-based QKD; thus, it has not yet been realized using practical quantum satellites.

In particular, regarding the link establishment problem, several studies have explored efficient schemes for entanglement distribution link establishment in quantum satellite networks [20–24]. In addition, Maule *et al.* [25] investigated the satellite scheduling strategies to establish secret keys among ground station pairs in a fair manner, where the E91 protocol [37] was adopted during the SGQL establishment within each minute-level time window. However, existing studies have mainly focused on the key generation efficiency while paying insufficient attention to key resource balance. Moreover, they have considered frequent link switching within each time window to improve link establishment efficiency, which may cause key negotiation interruptions due to device initialization and channel calibration.

Hence, the coordinated establishment of SGQLs and ISQLs incorporating more realistic constraints will be further explored in this work, in which single- and dual-layer LEO quantum satellite networks with different orbital inclinations will be considered, and the objectives of efficient key generation and balanced key resource distribution will be addressed.

3. PROBLEM FORMULATION

A. Network Structure

For LEO satellite networks, the Walker constellation [38,39] is a widely adopted satellite constellation structure, characterized by evenly spaced orbital longitudes and uniformly distributed satellites along each orbit. In particular, this constellation is well suited for supporting long-term global QKD services [29,30]. Based on orbital parameters, it adopts two typical configurations, i.e., Walker-star (polar-orbit) and Walker-delta (inclined-orbit), as illustrated in Fig. 1. Typically, Walker-delta constellations operate at inclinations of 30°–60° or 120°–150°, with a primary focus on mid-to-low latitudes and densely populated regions. Conversely, Walker-star constellations with an inclination of approximately 90° offer significant advantages in polar and high-latitude regions. However, due to fixed orbital altitudes and inclinations, a single-layer

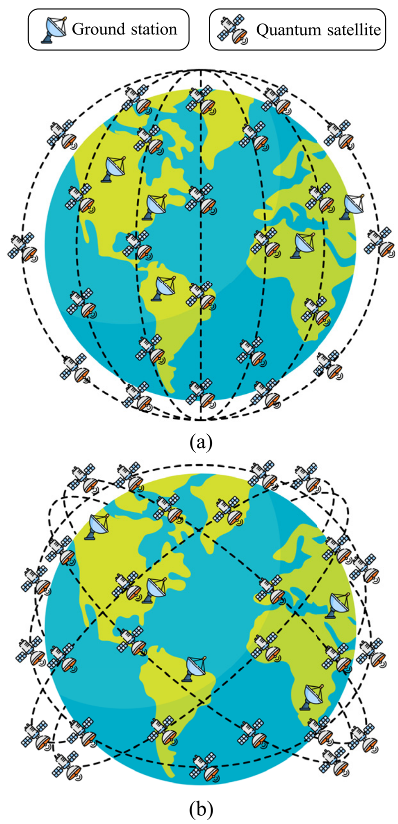


Fig. 1. Typical configurations of Walker constellations: (a) Walker-star constellation; (b) Walker-delta constellation.

Walker constellation may result in inadequate global coverage uniformity. Consequently, dual-layer or multi-layer constellations hold great potential for enhancing the overall performance of quantum satellite networks. For instance, in classical satellite communications, Starlink [17] also adopts a multi-layer constellation design to boost communication efficiency. To address the coverage imbalance of single-layer constellations, combining Walker-star and Walker-delta into a dual-layer LEO constellation with distinct inclinations can achieve stronger complementarity [16]. Additionally, satellites at different altitudes exhibit distinct transit times and coverage areas. Their mutual coordination can further expand the pool of candidate visible satellites available to ground stations, foster more effective topological cooperative relationships among satellites, and enhance the overall constellation robustness. Hence, both single-layer and dual-layer configurations of LEO quantum satellite networks are considered and compared in this work.

In single- or dual-layer LEO quantum satellite networks, multiple ground stations are deployed in conjunction with LEO quantum satellites. To facilitate key negotiation between quantum satellites and ground stations or between quantum satellites themselves, the QKD transmitter (QTx) at one node and the QKD receiver (QRx) at the other node need to be connected via a QKD link. Accordingly, each SGQL or ISQL is essentially established from the QTx to the QRx. Regarding the link direction between quantum satellites and ground stations, the downlink (i.e., from the satellite to the ground) exhibits reduced beam spreading and higher link efficiency compared to the uplink [9]. Therefore, ground stations are equipped with QRxs and establish SGQLs to the QTxs onboard quantum satellites. For ISQLs, each quantum satellite is equipped with a certain number of QTxs and QRxs. This means that each quantum satellite can act as either a sender or a receiver when establishing ISQLs in the LEO quantum satellite network.

For satellite-based QKD, the primary noise stems from background photons that enter the QRx alongside signal photons [9,15]. Consequently, QRxs face challenges in generating secret keys in the presence of sunlight, and daylight QKD has yet to be validated on practical quantum satellites. In contrast, the background light in the umbra zone (e.g., terrestrial night and the sun-unlit side in space) primarily originates from the Moon, stars, and human activities, with intensity orders of magnitude lower than that in sunlight [8]. In practice, the feasibility of nighttime QKD has been experimentally demonstrated by practical quantum satellites such as Micius [9,10] and Jinan-1 [11]. Thus, the illumination condition is regarded as a critical realistic constraint in this work. In particular, we consider three terrestrial/spatial illumination conditions, i.e., umbra (corresponding to terrestrial night and the sun-unlit side in space), penumbra (corresponding to terrestrial twilight and the partially illuminated intermediate zone between the sunlit and sun-unlit sides in space), and full solar illumination (corresponding to terrestrial daylight and the sunlit side in space).

The process of QKD link establishment in LEO quantum satellite networks is illustrated in Fig. 2. All quantum satellites and ground stations in Fig. 2 are assumed to be in the umbra zone. When a quantum satellite is visible to another quantum satellite or a ground station, this is indicated by a dashed line. Once a QKD link is established between the two nodes, it is denoted by a directional solid line. SGQLs and ISQLs are distinguished by line color. For example, when the quantum satellite s_1 transits over the ground station g_1, it can establish an SGQL with g_1, enabling continuous key generation until s_1 is no longer visible to g_1. Subsequently, the quantum satellite s_1 transits over the ground station g_3 and establishes an SGQL with g_3. The global key negotiation between geographically distant ground stations g_1 and g_3 can be achieved by leveraging the quantum satellite s_1 as a trusted relay [10]. Similarly, the ISQLs can be established based on the visibility between two quantum satellites. Notably, this work mainly focuses on the QKD link establishment process prior to global key negotiation.

In particular, when multiple ground stations and multiple quantum satellites are visible to a specific quantum satellite, QKD links cannot be established indefinitely. This is constrained by the limited number of QKD devices (i.e., QTxs and QRxs) onboard quantum satellites and the limited number of QRxs at ground stations. Let n_{st} and n_{sr} denote the number of QTxs and QRxs at each quantum satellite, respectively,

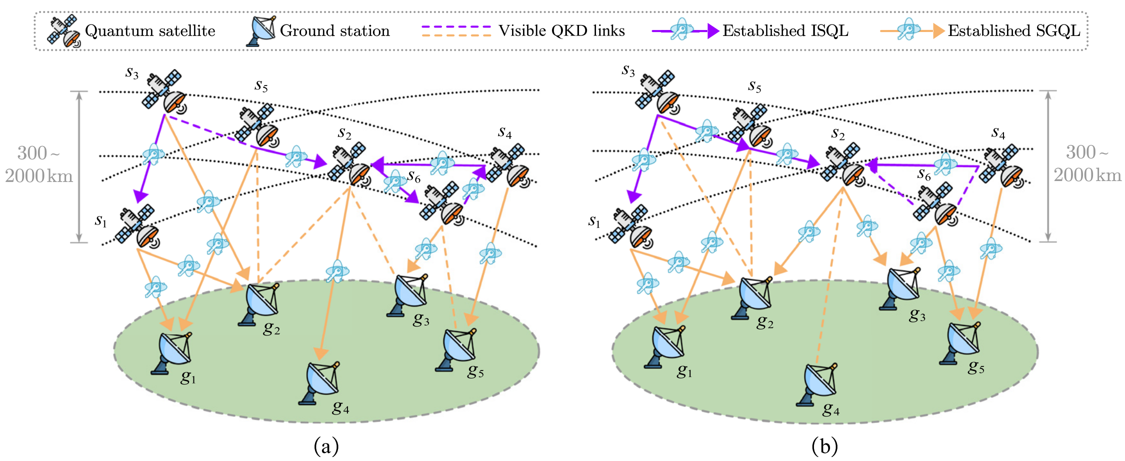


Fig. 2. Conventional QKD link establishment schemes in LEO quantum satellite networks: (a) RF, (b) FF.

as well as let n_{gr} denote the number of QRxs at each ground station. Under this QKD device quantity constraint, the QKD link establishment emerges as a critical problem.

Conventionally, the random fit (RF)-based and first fit (FF)-based QKD link establishment schemes are adopted. In the RF-based scheme, when the number of visible QKD links that can be established exceeds the QKD device quantity constraint of the corresponding quantum satellite or ground station, the QKD links are randomly selected for establishment under the limited number of QKD devices. For the FF-based scheme, all visible nodes (i.e., ground stations and quantum satellites) are numbered and subsequently selected in ascending order of their serial numbers. Furthermore, regarding the QKD link direction between two nodes in both schemes, the direction is chosen as downlink for SGQLs, while for ISQLs, it is randomly selected if a quantum satellite can act as both a sender and a receiver. The RF and FF schemes are exemplified in Figs. 2(a) and 2(b), respectively, where the QKD device quantities n_{st}, n_{sr}, and n_{gr} are all set to 2. As an example, for the ground station g_2, there are four visible QKD links, i.e., $s_1{-}g_2$, $s_2{-}g_2$, $s_3{-}g_2$, and $s_5{-}g_2$. In Fig. 2(a), two of these QKD links (i.e., $s_1{-}g_2$ and $s_3{-}g_2$) are randomly established, while in Fig. 2(b), two QKD links (i.e., $s_1{-}g_2$ and $s_2{-}g_2$) are selected sequentially based on the ascending serial numbers of quantum satellites.

It is important to note that different QKD link establishment schemes may result in variations in the number of generated secret keys at each node, thereby impacting the overall performance of LEO quantum satellite networks. The conventional schemes (i.e., RF and FF) merely perform simple selection from the full set of available QKD links, without considering key-related performance such as the key generation efficiency and balanced key resource distribution. This leads to inefficient and unbalanced satellite-based QKD operations while also reducing the success rate of QKD services.

B. Network Model

We model the LEO quantum satellite network as $G(V, E)$, where $V = V_g \cup V_s$ denotes the set of nodes (i.e., ground stations and quantum satellites), and $E = E^{SG} \cup E^{IS}$ denotes the set of visible QKD links. Specifically, V_g and V_s are the sets of ground stations and quantum satellites, respectively, whereas E^{SG} and E^{IS} are the sets of visible SGQLs and ISQLs, respectively. Let g_m ($g_m \in V_g$) represent the mth ground station, and let s_i ($s_i \in V_s$) and s_j ($s_j \in V_s$) represent the ith and jth quantum satellites, respectively. Additionally, $n_{st}(s_i)$ denotes the number of activated QTxs in the quantum satellite s_i, $n_{sr}(s_j)$ denotes the number of activated QRxs in the quantum satellite s_j, and $n_{gr}(g_m)$ denotes the number of activated QRxs in the ground station g_m.

Due to the high mobility of quantum satellites, the visible QKD links between nodes may be established or disconnected at any time, causing continuous changes to the topology of the LEO quantum satellite network. To facilitate tractable analysis, we discretize this time-varying process by capturing snapshots of network connections within a fixed interval τ. This interval refers to the duration of a time window, typically ranging from several seconds to a few minutes [20,22]. τ mainly influences the temporal granularity used to represent the quantum satellite network topology and link states. A smaller τ captures variations in topology and SKR more precisely, while a larger τ reduces the computational overhead but may degrade the accuracy of key-volume estimation. In this way, the dynamic LEO quantum satellite network is represented as a sequence of quasi-static topologies corresponding to discrete time windows. Let T denote the set of time windows, and let t ($t \in T$) denote the index of a time window. Specifically, a QKD link can only be established at time window t if its SKR (calculated in Subsection 3.C) is nonzero. Since QKD link establishment incurs calibration and initialization overheads, and the finite-key effect restricts the usability of extremely short connection durations, a single interval τ is generally insufficient for effective key generation. Moreover, given that the visible time window of each satellite-based QKD link is typically short, once a QKD link is established, it should not be disconnected (i.e., switched to another link) unless no secret key can be generated over that link. Hence, the selection of τ does not directly affect the link-switching performance considered in this work.

Within each time window, the state of each QKD link is fixed and can be characterized by a Boolean variable. These Boolean variables for an SGQL and an ISQL are denoted by $L_t(s_i, g_m)$ and $L_t(s_i, s_j)$, respectively. $L_t(s_i, g_m)$ is equal to 1 if the SGQL between quantum satellite s_i and ground station g_m is established within time window t, and 0 otherwise. $L_t(s_i, s_j)$ is equal to 1 if the ISQL between quantum satellites s_i and s_j is established within time window t, and 0 otherwise.

Different QKD link establishment schemes dictate the manner in which SGQLs and ISQLs are established. Following the establishment of QKD links, the quantum satellite availability emerges as a critical performance metric. Specifically, a quantum satellite serving as a trusted relay for global key negotiation is only available to a ground station or another quantum satellite provided that they have previously established a QKD link and shared secret keys via that link. For SGQLs, it directly reflects the coverage performance of quantum satellite constellations for ground stations. For ISQLs, it reflects the interconnectivity performance across quantum satellites in different orbits or layers. In this work, the quantum satellite availability is quantified by the proportion of quantum satellites utilized for QKD link establishment, which can be defined as follows:

$$A_{g_m}^{SG} = \frac{\sum_{s_i \in V_s} \mathrm{sgn}\left(\sum_{t=1}^{|T|} L_t(s_i, g_m)\right)}{|V_s|}, \tag{1}$$

$$A_{s_i}^{IS} = \frac{\sum_{s_j \in V_s} \mathrm{sgn}\left(\sum_{t=1}^{|T|} L_t(s_i, s_j)\right)}{|V_s| - 1}, \tag{2}$$

where $A_{g_m}^{SG}$ represents the quantum satellite availability via SGQLs for ground station g_m, and $A_{s_i}^{IS}$ represents the quantum satellite availability via ISQLs for quantum satellite s_i. Specifically, $\mathrm{sgn}(z)$ in Eqs. (1) and (2) is the sign function, which takes a value of 1 if $z \geq 1$, and 0 otherwise.

In addition to the quantum satellite availability, the number of QKD links established provides another perspective to characterize network connectivity. Notably, the process of establishing and subsequently releasing a QKD link may span multiple time windows. In this case, the same QKD link is counted only once across consecutive time windows. Specifically, under the link-switching constraint, the same QKD link will not be re-established within a short period, and the number of established QKD links is not counted repeatedly for each time window during a single transit of a quantum satellite over a ground station. Under identical visibility conditions, a larger total number of established QKD links indicates that a node may be connected to a broader set of visible nodes over the operational period, thereby implying richer connectivity and more candidate relay opportunities for subsequent end-to-end services. Notably, the total number of established QKD links is not intended to reflect the quality or amount of generated secret keys; instead, it serves as a supplementary metric for connection diversity and path flexibility in single- and dual-layer LEO quantum satellite networks. Accordingly, the total number of SGQLs established with ground station g_m is defined as

$$N_{g_m}^{\mathrm{SG}} = \sum_{s_i \in V_s} b(s_i, g_m), \tag{3}$$

where $b(s_i, g_m)$ denotes the total number of SGQLs established between quantum satellite s_i and ground station g_m over the full operation duration (i.e., encompassing all $|T|$ time windows). Similarly, the total number of ISQLs established with quantum satellite s_i is defined as

$$N_{s_i}^{\mathrm{IS}} = \sum_{s_j \in V_s} b(s_i, s_j), \tag{4}$$

where $b(s_i, s_j)$ represents the total number of ISQLs established between quantum satellites s_i and s_j over the full operation duration.

C. SKR Model

For QKD link establishment, we adopt the BB84 protocol, whose SKR is given by [40,41]

$$R = q \cdot \left\{ -Q_\mu \cdot f_e(E_\mu) \cdot H_2(E_\mu) + Q_1 \cdot [1 - H_2(e_1)] \right\}, \tag{5}$$

where q is the basis reconciliation factor, Q_μ is the overall gain of signal states, $f_e(\cdot)$ is the error correction efficiency, E_μ is the quantum bit error rate, Q_1 is the gain of single-photon pulses, e_1 is the error rate of single-photon pulses, and $H_2(\cdot)$ is the binary Shannon entropy function. Furthermore, the standard decoy-state method [41,42] is commonly utilized in conjunction with the BB84 protocol, which has been validated on practical quantum satellites such as Micius [9,10] and Jinan-1 [11]. Let $R_t^{\mathrm{SG}}(s_i, g_m)$ represent the SKR of the SGQL between quantum satellite s_i and ground station g_m at time window t, and let $R_t^{\mathrm{IS}}(s_i, s_j)$ represent the SKR of the ISQL between quantum satellites s_i and s_j at time window t.

Due to the relatively short duration of a single satellite pass and the time required to accumulate sufficient keys for key distillation under the finite-key effect, key distillation timeliness becomes a critical challenge. For the Micius quantum satellite, the key distillation timeliness is 2–3 days [9], whereas for the Jinan-1 quantum satellite, key distillation is performed in real time [11]. Hence, in this work, the finite-key effect is not directly incorporated into the SKR model, but we adopt relevant parameters from Jinan-1 to facilitate real-time QKD and introduce a link-switching constraint to avoid frequent link switching.

Furthermore, the total transmission efficiency η and the background yield Y_0 are two critical parameters affecting the SKR. Specifically, η is modeled by estimating channel losses and receiver-side component losses, as given by [43]

$$\eta = \eta_{\mathrm{diff}} \cdot \eta_{\mathrm{atm}} \cdot \eta_{\mathrm{turb}} \cdot \eta_{\mathrm{pde}} \cdot \eta_{\mathrm{rx}} \cdot \eta_{\mathrm{cpl}} \cdot \eta_{\mathrm{pat}}, \tag{6}$$

$$\eta_{\mathrm{diff}} = \frac{(D_{\mathrm{r}} \cdot \eta_{\mathrm{obs}})^2}{(\theta \cdot d)^2}, \tag{7}$$

$$\eta_{\mathrm{atm}} = \eta_\lambda^{\csc(\alpha)}, \tag{8}$$

where η_{diff} is the diffraction-induced efficiency, η_{atm} is the atmospheric transmissivity, η_{turb} is the beam decoherence efficiency due to turbulence effects, η_{pde} is the photon detection efficiency, η_{rx} is the receiver optical efficiency, η_{cpl} is the coupling efficiency, and η_{pat} is the pointing and tracking efficiency. Specifically, η_{diff} is given by Eq. (7), where D_{r} is the diameter of the receiver telescope, η_{obs} denotes the transmissivity of the receiver telescope due to the central obstruction, θ is the full-angle beam divergence, and d is the link distance (between a quantum satellite and a ground station, or between two quantum satellites). η_{atm} is given by Eq. (8), where η_λ is the atmospheric transmissivity at zenith and varies with wavelength λ, and α is the satellite elevation angle. For SGQLs, all factors defined in Eq. (6) need to be considered to characterize the satellite–ground link transmission efficiency. For ISQLs, by contrast, η_{atm} and η_{turb} do not exist since ISQLs operate in a vacuum environment, while other factors, including η_{diff}, η_{pde}, η_{rx}, η_{cpl}, and η_{pat}, are still required to characterize the inter-satellite link transmission efficiency. Given the high relative speed between a quantum satellite and a ground station, as well as between two quantum satellites, the influencing factors considered in Eq. (6) incorporate several practical parameters such as the pointing and tracking efficiency η_{pat}. Furthermore, the feasibility of establishing wireless inter-satellite optical links has been demonstrated in the Starlink system [44].

Under different illumination conditions, the background yield Y_0 is influenced by the dark counts and background noise, which is given by [15,45]

$$Y_0 = \eta_{\mathrm{pde}} \cdot \eta_{\mathrm{rx}} \cdot \eta_{\mathrm{cpl}} \cdot \frac{H_{\mathrm{b}} \cdot \Omega_{\mathrm{FOV}} \cdot \pi \cdot R_{\mathrm{rec}}^2 \cdot \lambda \cdot \Delta\lambda \cdot \Delta t}{\mathrm{h} \cdot \mathrm{c}} + f_{\mathrm{dark}} \cdot \Delta t, \tag{9}$$

where f_{dark} is the overall dark count rate, Δt is the detection time window, H_{b} is the background radiance, Ω_{FOV} is the solid angle field of view (FOV), R_{rec} is the radius of the receiver telescope, $\Delta\lambda$ is the filter bandwidth, h is the Planck constant, and c is the speed of light. It is important to note that the value of H_{b} varies for the three terrestrial/spatial illumination conditions.

Overall, the difference between the SKR models for SGQLs and ISQLs lies in the total transmission efficiency η defined in

Eq. (6), which is a pivotal parameter affecting the SKR value calculated in Eq. (5). The general SKR formula in Eq. (5) and the background yield in Eq. (9) are applicable to both types of QKD links, although the values of some parameters differ between SGQLs and ISQLs.

D. SKV Model

Due to the high dynamic nature of LEO quantum satellite networks, it is challenging to ensure a continuous key supply via dynamic QKD links. In this regard, the construction of a key pool [5] can enhance the efficiency of key resources by storing the generated secret keys for subsequent extraction. Since the node positions are fixed within each time window, the corresponding SKR remains constant therein. Typically, the rectangular integral method [14] can be utilized to calculate the GKV. However, the SKR for a satellite overpass initially increases until the satellite elevation angle reaches 90° and then declines until the satellite is no longer visible [9,43]. In this context, the rectangular integral method may yield imprecise results when estimating the GKV if the interval of each time window is relatively large (i.e., 1 min). Hence, we adopt a trapezoidal calculation method to estimate the GKV within time window t. The GKV of the SGQL (s_i, g_m) within time window t is estimated as

$$\Delta K_t^{\mathrm{SG}}(s_i, g_m) = \frac{\left[R_t^{\mathrm{SG}}(s_i, g_m) + R_{t+1}^{\mathrm{SG}}(s_i, g_m)\right] \cdot \tau}{2}. \quad \mathbf{(10)}$$

Similarly, the GKV of the ISQL (s_i, s_j) within time window t is estimated as

$$\Delta K_t^{\mathrm{IS}}(s_i, s_j) = \frac{\left[R_t^{\mathrm{IS}}(s_i, s_j) + R_{t+1}^{\mathrm{IS}}(s_i, s_j)\right] \cdot \tau}{2}. \quad \mathbf{(11)}$$

For each time window t, the RKV in the key pool associated with the SGQL or ISQL is defined as the cumulative key volume generated from all time windows prior to t. Accordingly, the RKV for the SGQL (s_i, g_m) at time window t is expressed as

$$P_t^{\mathrm{SG}}(s_i, g_m) = \sum_{t'=1}^{t-1} \Delta K_{t'}^{\mathrm{SG}}(s_i, g_m) \cdot x_{t'}^{\mathrm{SG}}(s_i, g_m), \quad \mathbf{(12)}$$

where the Boolean variable $x_{t'}^{\mathrm{SG}}(s_i, g_m)$ indicates whether quantum satellite s_i (acting as the QTx) establishes an SGQL with ground station g_m (acting as the QRx) at time window t'. $x_{t'}^{\mathrm{SG}}(s_i, g_m)$ takes a value of 1 if the SGQL (s_i, g_m) is established, and 0 otherwise. Similarly, the RKV for the ISQL (s_i, s_j) at time window t is expressed as

$$P_t^{\mathrm{IS}}(s_i, s_j) = \sum_{t'=1}^{t-1} \Delta K_{t'}^{\mathrm{IS}}(s_i, s_j) \cdot x_{t'}^{\mathrm{IS}}(s_i, s_j), \quad \mathbf{(13)}$$

where the Boolean variable $x_{t'}^{\mathrm{IS}}(s_i, s_j)$ indicates whether quantum satellite s_i (acting as the QTx) establishes an ISQL with quantum satellite s_j (acting as the QRx) at time window t'. $x_{t'}^{\mathrm{IS}}(s_i, s_j)$ takes a value of 1 if the ISQL (s_i, s_j) is established, and 0 otherwise.

4. INTEGER LINEAR PROGRAMMING MODELS

In this section, we develop two ILP models (i.e., ILP-G and ILP-R) to obtain optimal solutions for QKD link establishment in LEO quantum satellite networks.

A. Objectives

The LEO quantum satellite network is regarded as a general infrastructure supporting multiple types of security-required services, rather than only end-to-end services between ground stations. For instance, emerging inter-satellite services have become increasingly common in satellite networks, such as on-board federated learning [46] and observation task scheduling [47]. For such inter-satellite services, ISQLs can be employed to ensure secure data transmission between quantum satellites, and their performance is also significant. Hence, to consider a more general scenario, we treat SGQLs and ISQLs equally in this work. For QKD link establishment, we aim either to select QKD links with higher GKV to maximize the total generated key resources, or to prioritize QKD links with lower RKV to achieve a relatively balanced distribution of key resources. Notably, since service traffic is not considered during QKD link establishment, no generated keys are consumed, allowing the RKV to accumulate over time. Accordingly, the objectives of the ILP-G and ILP-R models are to maximize GKV and to prioritize QKD links with the minimum RKV over the entire quantum satellite network, respectively. To address this task, we assign distinct link weights to SGQLs and ISQLs. In the ILP-G model, the link weight is defined as the GKV of the corresponding QKD link; while in the ILP-R model, it is defined as the difference between the current network-wide maximum RKV and the RKV for the corresponding QKD link. The link weight $w_t^{\mathrm{SG}}(s_i, g_m)$ of SGQL (s_i, g_m) and the link weight $w_t^{\mathrm{IS}}(s_i, s_j)$ of ISQL (s_i, s_j) at time window t are defined as follows:

$$w_t^{\mathrm{SG}}(s_i, g_m) = \begin{cases} \Delta K_t^{\mathrm{SG}}(s_i, g_m) & \text{for ILP-G} \\ P_t^{\max} - P_t^{\mathrm{SG}}(s_i, g_m) & \text{for ILP-R} \end{cases}, \quad \mathbf{(14)}$$

$$w_t^{\mathrm{IS}}(s_i, s_j) = \begin{cases} \Delta K_t^{\mathrm{IS}}(s_i, s_j) & \text{for ILP-G} \\ P_t^{\max} - P_t^{\mathrm{IS}}(s_i, s_j) & \text{for ILP-R} \end{cases}, \quad \mathbf{(15)}$$

where $P_t^{\max} = \max\left\{ \max_{e \in E^{\mathrm{SG}}} \left[P_t^{\mathrm{SG}}(e)\right], \max_{e \in E^{\mathrm{IS}}} \left[P_t^{\mathrm{IS}}(e)\right] \right\}$. $P_t^{\max}$ denotes the maximum RKV value across all QKD links in the entire network at time window t. When prioritizing the establishment of QKD links with the smallest RKV, one cannot simply set the link weight to RKV and minimize it directly, as this would lead the ILP model to avoid establishing any links to achieve an optimal solution. To resolve this issue, we introduce the parameter $P_t^{\max}$ and use the difference between $P_t^{\max}$ and the RKV of the current link as the link weight. This difference assigns a larger weight to links with a smaller RKV. By maximizing this weight over the entire quantum satellite network, the ILP-R model prioritizes QKD links with the smallest RKV, which is analogous to how links with the largest GKV are prioritized in ILP-G.

For each time window t, the objective functions of the ILP-G and ILP-R models are defined as

$$\text{Maximize} \sum_{(s_i,g_m)\in E^{\text{SG}}} w_t^{\text{SG}}(s_i, g_m)\cdot x_t^{\text{SG}}(s_i, g_m) + \sum_{(s_i,s_j)\in E^{\text{IS}}} w_t^{\text{IS}}(s_i, s_j)\cdot x_t^{\text{IS}}(s_i, s_j), \quad \textbf{(16)}$$

where $x_t^{\text{SG}}(s_i, g_m)$ and $x_t^{\text{IS}}(s_i, s_j)$ are Boolean variables as defined earlier. In addition, the ILP-G and ILP-R models share the same constraints as follows.

B. Constraints

1) QKD device constraint:

$$\sum_{g_m\in V_{\text{g}}} x_t^{\text{SG}}(s_i, g_m) + \sum_{s_j\in V_{\text{s}}} x_t^{\text{IS}}(s_i, s_j) \le n_{\text{st}}, \ \forall s_i \in V_{\text{s}},\ t\in T, \quad \textbf{(17)}$$

$$\sum_{s_i\in V_{\text{s}}} x_t^{\text{SG}}(s_i, g_m) \le n_{\text{gr}}, \ \forall g_m \in V_{\text{g}},\ t\in T, \quad \textbf{(18)}$$

$$\sum_{s_i\in V_{\text{s}}} x_t^{\text{IS}}(s_i, s_j) \le n_{\text{sr}}, \ \forall s_j \in V_{\text{s}},\ t\in T. \quad \textbf{(19)}$$

Given the limited QKD device capacity of ground stations and quantum satellites, Eq. (17) ensures that the maximum number of QTxs per quantum satellite does not exceed n_{st}, while Eqs. (18) and (19) ensure that the maximum numbers of QRxs per ground station and per quantum satellite do not exceed n_{gr} and n_{sr}, respectively.

2) SGQL constraint:

$$\sum_{s_i\in V_{\text{s}}} x_t^{\text{SG}}(g_m, s_i) = 0, \ \forall g_m \in V_{\text{g}},\ t\in T. \quad \textbf{(20)}$$

For SGQLs, only the downlink is utilized for QKD link establishment. Eq. (20) ensures that a ground station cannot act as the sender.

3) ISQL constraint:

$$x_t^{\text{IS}}(s_i, s_j) + x_t^{\text{IS}}(s_j, s_i) \le 1, \ \forall s_i \in V_{\text{s}},\ s_j \in V_{\text{s}},\ t\in T. \quad \textbf{(21)}$$

To avoid redundant ISQL establishment under the limited number of QKD devices, Eq. (21) ensures that only one unidirectional ISQL (either $s_i \to s_j$ or $s_j \to s_i$) can exist between quantum satellites s_i and s_j.

4) Link-switching constraint:

$$x_{t+1}^{\text{SG}}(s_i, g_m) \ge x_t^{\text{SG}}(s_i, g_m)\cdot \text{sgn}\left(\Delta K_{t+1}^{\text{SG}}(s_i, g_m)\right), \quad \forall s_i \in V_{\text{s}},\ g_m \in V_{\text{g}},\ t\in T, \quad \textbf{(22)}$$

$$x_{t+1}^{\text{IS}}(s_i, s_j) \ge x_t^{\text{IS}}(s_i, s_j)\cdot \text{sgn}\left(\Delta K_{t+1}^{\text{IS}}(s_i, s_j)\right), \quad \forall s_i \in V_{\text{s}},\ s_j \in V_{\text{s}},\ t\in T. \quad \textbf{(23)}$$

Establishing QKD links incurs calibration and initialization overheads, and the finite-key effect also constrains the effective key generation time. In practice, frequent link switching may reduce the overall key generation efficiency. Hence, established QKD links should be maintained unless no secret key can be generated over them, which is consistent with the state-of-the-art practical quantum satellite platform [11]. Equations (22) and (23) ensure that, after an SGQL and an ISQL (respectively corresponding to the two equations) are established within a given time window, the connection is maintained within the subsequent time window as long as the QKD link remains capable of generating secret keys.

5. ADAPTIVE QKD LINK ESTABLISHMENT SCHEMES AND ALGORITHMS

In this section, we devise three adaptive QKD link establishment schemes, namely, the MaxKR, MaxGKV, and MinRKV schemes. To implement these schemes, three heuristic algorithms are designed.

The design objective of MaxKR is to fully leverage QKD links with higher SKR to generate more secret keys. In LEO quantum satellite networks, the channel loss and noise levels of SGQLs and ISQLs vary rapidly with distance, elevation angle, and illumination conditions, resulting in substantial SKR discrepancies across different QKD links. Therefore, by sorting all candidate QKD links in descending order of SKR at each time window, the MaxKR scheme adaptively prioritizes the establishment of top-SKR candidate QKD links. Additionally, if a given node can act as both a QRx and a QTx, the communication direction is randomly selected. Figure 3 illustrates the QKD links established via MaxKR under the following constraints: one QTx and one QRx per quantum satellite and one QRx per ground station. Quantum satellites s_1 and s_2 pass over ground station g_1; s_2 and s_3 pass over ground station g_2; s_3 and s_4 pass over ground station g_3. There are six visible SGQLs in total, plus three visible ISQLs, all of which are identified as candidate QKD links. These candidate QKD links are sorted in descending order of SKRs, with values of 18 kbps, 12 kbps, 10 kbps, 8 kbps, etc. The MaxKR scheme then sequentially establishes the corresponding QKD links: (s_3, s_4), (s_4, g_3), (s_1, s_2), and (s_2, g_2). For an ISQL, e.g., (s_3, s_4), the MaxKR scheme randomly selects the direction $s_3 \to s_4$ to establish the link. For SGQLs, only the downlink is employed. The remaining candidate QKD links cannot be established, as insufficient QRx/QTx resources are available.

The design objective of MaxGKV is to consider the cumulative key generation contribution of QKD links within a time window, rather than merely their instantaneous performance.

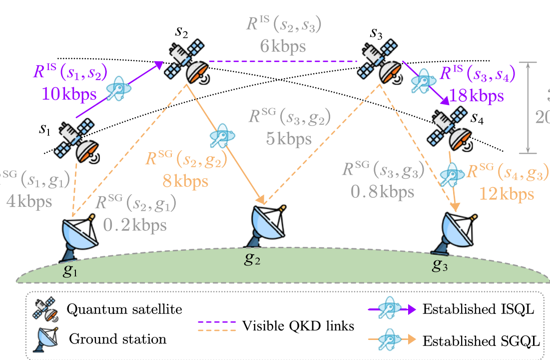


Fig. 3. Illustration of the MaxKR-based QKD link establishment scheme.

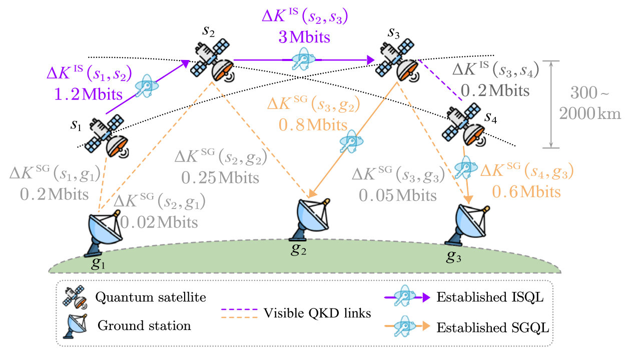


Fig. 4. Illustration of the MaxGKV-based QKD link establishment scheme.

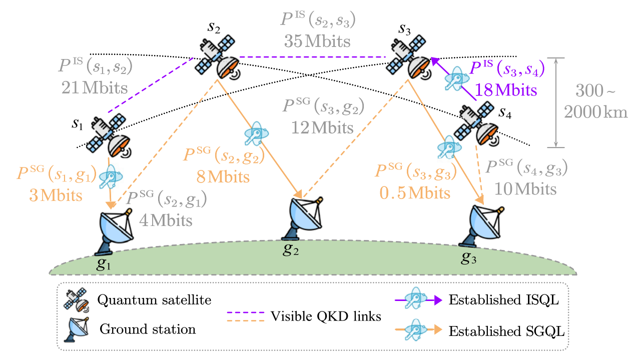


Fig. 5. Illustration of the MinRKV-based QKD link establishment scheme.

Specifically, the actual GKV of a candidate QKD link depends not only on its SKR but also on the duration of its continuous operational availability. For example, a link with moderate SKR yet capable of maintaining operation for an extended period may achieve a higher cumulative GKV than a link with extremely high SKR but a very short operational duration. Therefore, the MaxGKV scheme adaptively prioritizes establishing top-GKV candidate QKD links within a time window by calculating the GKV of each candidate QKD link and sorting them in descending order. Figure 4 exemplifies the QKD links established via MaxGKV under conditions similar to those in Fig. 3. There are six visible SGQLs and three visible ISQLs in total. These candidate QKD links are sorted in descending order of GKV, with values of 3 Mbits, 1.2 Mbits, 0.8 Mbits, 0.6 Mbits, etc. The MaxGKV scheme then sequentially establishes the corresponding QKD links: (s_2, s_3), (s_1, s_2), (s_3, g_2), and (s_4, g_3). For an ISQL, e.g., (s_2, s_3), the MaxGKV scheme randomly selects the direction $s_2 \rightarrow s_3$ to establish the link. Similarly, only the downlink is utilized for SGQLs.

For practical QKD services, the balance of key resources is a critical indicator for improving service success rates. The MaxKR and MaxGKV schemes generally prioritize the quantity of key resources, while the long-term neglect of relatively underprivileged links results in key resource imbalance across the quantum satellite network. This imbalance may reduce the success rate of QKD services. Therefore, to mitigate this problem, the MinRKV scheme ranks the RKV of all candidate QKD links at each time window and adaptively prioritizes establishing links with the lowest RKV. Figure 5 illustrates the QKD links established via MinRKV under conditions similar to those in Fig. 3. There are six visible SGQLs and three visible ISQLs in total. These candidate QKD links are sorted in ascending order of RKV, with values of 0.5 Mbits, 3 Mbits, 4 Mbits, 8 Mbits, etc. The MinRKV scheme then sequentially establishes the corresponding QKD links: (s_3, g_3), (s_1, g_1), (s_2, g_2), and (s_4, s_3). For an ISQL, (s_4, s_3), the MinRKV scheme randomly selects the direction $s_4 \rightarrow s_3$ to establish the link. For SGQLs, only the downlink is adopted. Specifically, although the RKV of SGQLs (s_2, g_1), (s_4, g_3), and (s_3, g_2) is lower than that of the established ISQL (s_4, s_3), these links cannot be further established due to insufficient QTx or QRx resources at their corresponding nodes.

In LEO quantum satellite networks, the coordinated establishment of SGQLs and ISQLs constitutes a spatiotemporal coupled optimization problem. To optimize the overall performance of the quantum satellite network, the link establishment schemes are designed mainly based on the SKR, GKV, and RKV metrics of each QKD link, with no further distinction between SGQLs and ISQLs by link type. Although the ILP model can, in principle, yield globally optimal solutions, its computational complexity and solution latency become prohibitive in large-scale topologies with fine-grained dynamics, thereby limiting its applicability for real-time planning. To improve computational efficiency while retaining high solution quality, we propose the three aforementioned QKD link establishment schemes and design their corresponding heuristic algorithms in the following subsections.

A. Algorithm 1: MaxKR-Based QKD Link Establishment Algorithm

Algorithm 1 presents the details of the MaxKR-based QKD link establishment algorithm. At the beginning, line 1 initializes the connection status of all visible QKD links. For each time window, lines 3–22 compute the SKR (handled in lines 5 and 13) and maintain previously established QKD links until they can no longer generate secret keys (handled in lines 6–10 and 14–18). Additionally, lines 8 and 16 update the activated QTxs and QRxs in the corresponding quantum satellite and ground station, while lines 9 and 17 set the SKR of the QKD link (that needs to remain connected) to zero. This prevents the link from being reselected in subsequent operations. Lines 19–21 randomly select one of the two ISQLs from (s_i, s_j) and (s_j, s_i) (for which both SKRs are greater than zero) and set the SKR of the other ISQL to zero, which avoids establishing bi-directional ISQLs between the same pair of nodes. Line 23 selects all candidate SGQLs and ISQLs with non-zero SKRs and sorts them in descending order by SKR. Lines 24–43 prioritize establishing QKD links with the largest SKR while satisfying the QKD device quantity constraints for each node, and this process continues until all SKR-sorted QKD links have been traversed.

Algorithm 1. MaxKR-Based QKD Link Establishment Algorithm

Input: $G(V, E)$, n_{st}, n_{sr}, n_{gr}, τ, T.
Output: $x_t^{\mathrm{SG}}(s_i, g_m)$ for SGQLs and $x_t^{\mathrm{IS}}(s_i, s_j)$ for ISQLs at each time window $t \in T$.
1: initialize $x_0^{\mathrm{SG}}(s_i, g_m)$, $x_0^{\mathrm{IS}}(s_i, s_j) \leftarrow 0$ for all QKD links;
2: **for** each time window $t \in T$ **do**
3: initialize $x_t^{\mathrm{SG}}(s_i, g_m)$, $x_t^{\mathrm{IS}}(s_i, s_j) \leftarrow 0$ for all QKD links and $n_{\mathrm{st}}(s_i)$, $n_{\mathrm{sr}}(s_j)$, $n_{\mathrm{gr}}(g_m) \leftarrow 0$ for all nodes;
4: **for** each $(s_i, g_m) \in E^{\mathrm{SG}}$ **do**
5: calculate $R_t^{\mathrm{SG}}(s_i, g_m)$ according to Eqs. (5)–(9);
6: **if** $x_{t-1}^{\mathrm{SG}}(s_i, g_m) = 1$ and $R_t^{\mathrm{SG}}(s_i, g_m) > 0$ **then**
7: $x_t^{\mathrm{SG}}(s_i, g_m) \leftarrow 1$;
8: $n_{\mathrm{st}}(s_i) \leftarrow n_{\mathrm{st}}(s_i) + 1$, $n_{\mathrm{gr}}(g_m) \leftarrow n_{\mathrm{gr}}(g_m) + 1$;
9: $R_t^{\mathrm{SG}}(s_i, g_m) \leftarrow 0$;
10: **end if**
11: **end for**
12: **for** each $(s_i, s_j) \in E^{\mathrm{IS}}$ **do**
13: calculate $R_t^{\mathrm{IS}}(s_i, s_j)$ according to Eqs. (5)–(9);
14: **if** $x_{t-1}^{\mathrm{IS}}(s_i, s_j) = 1$ and $R_t^{\mathrm{IS}}(s_i, s_j) > 0$ **then**
15: $x_t^{\mathrm{IS}}(s_i, s_j) \leftarrow 1$;
16: $n_{\mathrm{st}}(s_i) \leftarrow n_{\mathrm{st}}(s_i) + 1$, $n_{\mathrm{sr}}(s_j) \leftarrow n_{\mathrm{sr}}(s_j) + 1$;
17: $R_t^{\mathrm{IS}}(s_i, s_j) \leftarrow 0$, $R_t^{\mathrm{IS}}(s_j, s_i) \leftarrow 0$;
18: **end if**
19: **if** $R_t^{\mathrm{IS}}(s_i, s_j) > 0$ and $R_t^{\mathrm{IS}}(s_j, s_i) > 0$ **then**
20: randomly select one of the two ISQLs from (s_i, s_j) and (s_j, s_i), and set the SKR of the other ISQL to zero;
21: **end if**
22: **end for**
23: insert the SGQLs and ISQLs with SKR>0 into the candidate QKD link set E', and sort them in descending order by SKR;
24: **for** each candidate QKD link $e \in E'$ **do**
25: **if** $e \in E^{\mathrm{SG}}$ **then**
26: determine s_i and g_m for SGQL e;
27: **if** $n_{\mathrm{st}}(s_i) = n_{\mathrm{st}}$ or $n_{\mathrm{gr}}(g_m) = n_{\mathrm{gr}}$ **then**
28: **continue**;
29: **else**
30: establish SGQL (s_i, g_m): $x_t^{\mathrm{SG}}(s_i, g_m) \leftarrow 1$;
31: $n_{\mathrm{st}}(s_i) \leftarrow n_{\mathrm{st}}(s_i) + 1$, $n_{\mathrm{gr}}(g_m) \leftarrow n_{\mathrm{gr}}(g_m) + 1$;
32: **end if**
33: **end if**
34: **if** $e \in E^{\mathrm{IS}}$ **then**
35: determine s_i and s_j for ISQL e;
36: **if** $n_{\mathrm{st}}(s_i) = n_{\mathrm{st}}$ or $n_{\mathrm{sr}}(s_j) = n_{\mathrm{sr}}$ **then**
37: **continue**;
38: **else**
39: establish ISQL (s_i, s_j): $x_t^{\mathrm{IS}}(s_i, s_j) \leftarrow 1$;
40: $n_{\mathrm{st}}(s_i) \leftarrow n_{\mathrm{st}}(s_i) + 1$, $n_{\mathrm{sr}}(s_j) \leftarrow n_{\mathrm{sr}}(s_j) + 1$;
41: **end if**
42: **end if**
43: **end for**
44: **end for**
45: **return** $x_t^{\mathrm{SG}}(s_i, g_m)$ for SGQLs and $x_t^{\mathrm{IS}}(s_i, s_j)$ for ISQLs at each time window $t \in T$.

B. Algorithm 2: MaxGKV-Based QKD Link Establishment Algorithm

Algorithm 2 presents the details of the MaxGKV-based QKD link establishment algorithm. Line 1 initializes the connection variables for all visible QKD links. For each time window,

Algorithm 2. MaxGKV-Based QKD Link Establishment Algorithm

Input: $G(V, E)$, n_{st}, n_{sr}, n_{gr}, τ, T.
Output: $x_t^{\mathrm{SG}}(s_i, g_m)$ for SGQLs and $x_t^{\mathrm{IS}}(s_i, s_j)$ for ISQLs at each time window $t \in T$.
1: initialize $x_0^{\mathrm{SG}}(s_i, g_m)$, $x_0^{\mathrm{IS}}(s_i, s_j) \leftarrow 0$ for all QKD links;
2: **for** each time window $t \in T$ **do**
3: initialize $x_t^{\mathrm{SG}}(s_i, g_m)$, $x_t^{\mathrm{IS}}(s_i, s_j) \leftarrow 0$ for all QKD links and $n_{\mathrm{st}}(s_i)$, $n_{\mathrm{sr}}(s_j)$, $n_{\mathrm{gr}}(g_m) \leftarrow 0$ for all nodes;
4: **for** each $(s_i, g_m) \in E^{\mathrm{SG}}$ **do**
5: calculate $R_t^{\mathrm{SG}}(s_i, g_m)$ and $R_{t+1}^{\mathrm{SG}}(s_i, g_m)$ according to Eqs. (5)–(9);
6: $\Delta K_t^{\mathrm{SG}}(s_i, g_m) \leftarrow \left[R_t^{\mathrm{SG}}(s_i, g_m) + R_{t+1}^{\mathrm{SG}}(s_i, g_m)\right] \cdot \tau/2$;
7: call lines 6 to 10 in Algorithm 1;
8: **end for**
9: **for** each $(s_i, s_j) \in E^{\mathrm{IS}}$ **do**
10: calculate $R_t^{\mathrm{IS}}(s_i, s_j)$ and $R_{t+1}^{\mathrm{IS}}(s_i, s_j)$ according to Eqs. (5)–(9);
11: $\Delta K_t^{\mathrm{IS}}(s_i, s_j) \leftarrow \left[R_t^{\mathrm{IS}}(s_i, s_j) + R_{t+1}^{\mathrm{IS}}(s_i, s_j)\right] \cdot \tau/2$;
12: call lines 14 to 21 in Algorithm 1;
13: **end for**
14: insert the SGQLs and ISQLs with SKR > 0 into the candidate QKD link set E', and sort them in descending order by GKV;
15: call lines 24 to 43 in Algorithm 1;
16: **end for**
17: **return** $x_t^{\mathrm{SG}}(s_i, g_m)$ for SGQLs and $x_t^{\mathrm{IS}}(s_i, s_j)$ for ISQLs at each time window $t \in T$.

lines 3–13 compute the GKV and maintain existing connections, where lines 6 and 11 calculate the GKV of SGQLs and ISQLs, respectively, and lines 7 and 12 enforce the link-switching constraints (consistent with those in Algorithm 1). Line 14 inserts SGQLs and ISQLs with non-zero GKV into the candidate QKD link set and sorts these links in descending order by GKV. Line 15 calls the corresponding steps from Algorithm 1, which prioritizes establishing QKD links with the largest GKV while satisfying the QKD device quantity constraints for each node, and this process continues until all GKV-sorted QKD links have been traversed.

C. Algorithm 3: MinRKV-Based QKD Link Establishment Algorithm

Algorithm 3 presents the details of the MinRKV-based QKD link establishment algorithm. Line 1 initializes the connection variables for all visible QKD links. At the initial time window ($t = 1$), no QKD links have been established yet, hence all links have an RKV of 0. This makes it not feasible to establish QKD links based on the RKV. Thus, lines 3–4 treat this initial time window as a degenerate case, establishing the initial QKD links by calling the corresponding steps in Algorithm 2 and beginning to accumulate secret keys. For each subsequent time window ($t > 1$), lines 5–16 compute the RKV and maintain existing connections, where lines 9 and 14 compute the RKV of SGQLs and ISQLs, respectively, according to Eqs. (12) and (13). In addition, lines 10 and 15 enforce the link-switching constraints consistent with those in Algorithm 1. Line 17 inserts SGQLs and ISQLs with non-zero SKR into the candidate QKD link set and sorts them in ascending order by RKV.

Algorithm 3. MinRKV-Based QKD Link Establishment Algorithm

Input: $G(V, E)$, n_{st}, n_{sr}, n_{gr}, τ, T.
Output: $x_t^{\mathrm{SG}}(s_i, g_m)$ for SGQLs and $x_t^{\mathrm{IS}}(s_i, s_j)$ for ISQLs at each time window $t \in T$.
1: initialize $x_0^{\mathrm{SG}}(s_i, g_m)$, $x_0^{\mathrm{IS}}(s_i, s_j) \leftarrow 0$ for all QKD links;
2: **for** each time window $t \in T$ **do**
3: **if** time window $t = 1$ **then**
4: call lines 3 to 15 in Algorithm 2;
5: **else**
6: initialize $x_t^{\mathrm{SG}}(s_i, g_m)$, $x_t^{\mathrm{IS}}(s_i, s_j) \leftarrow 0$ for all QKD links and $n_{\mathrm{st}}(s_i)$, $n_{\mathrm{sr}}(s_j)$, $n_{\mathrm{gr}}(g_m) \leftarrow 0$ for all nodes;
7: **for** each $(s_i, g_m) \in E^{\mathrm{SG}}$ **do**
8: calculate $R_t^{\mathrm{SG}}(s_i, g_m)$ according to Eqs. (5)–(9);
9: calculate $P_t^{\mathrm{SG}}(s_i, g_m)$ according to Eq. (12);
10: call lines 6 to 10 in Algorithm 1;
11: **end for**
12: **for** each $(s_i, s_j) \in E^{\mathrm{IS}}$ **do**
13: calculate $R_t^{\mathrm{IS}}(s_i, s_j)$ according to Eqs. (5)–(9);
14: calculate $P_t^{\mathrm{IS}}(s_i, s_j)$ according to Eq. (13);
15: call lines 14 to 21 in Algorithm 1;
16: **end for**
17: insert the SGQLs and ISQLs with SKR > 0 into the candidate QKD link set E', and sort them in ascending order by RKV;
18: call lines 24 to 43 in Algorithm 1;
19: **end if**
20: **end for**
21: **return** $x_t^{\mathrm{SG}}(s_i, g_m)$ for SGQLs and $x_t^{\mathrm{IS}}(s_i, s_j)$ for ISQLs at each time window $t \in T$.

Similar to Algorithm 1, line 18 prioritizes establishing QKD links with the smallest RKV while following the constraints on the number of QTxs and QRxs at each node, and this process continues until all RKV-sorted QKD links have been traversed.

Furthermore, Algorithms 1–3 share similar operational logic; hence, their worst-case time complexities are analogous. For each time window $t \in T$, the time complexity of traversing all SGQLs and ISQLs is $O(|E^{\mathrm{SG}}| + |E^{\mathrm{IS}}|) = O(|E|)$. The time complexity of sorting all candidate QKD links is $O(|E| \log |E|)$. Thus, the overall time complexity of these three heuristic algorithms is $O(|T|(|E| + |E| \log |E|))$.

6. EVALUATION AND ANALYSIS

In this section, the ILP-G and ILP-R models, along with the MaxKR-, MaxGKV-, and MinRKV-based heuristic algorithms for QKD link establishment, are evaluated in single- and dual-layer LEO quantum satellite networks via numerical simulations. Two quantum satellite constellations are adopted, including a single-layer LEO quantum satellite constellation and a dual-layer LEO quantum satellite constellation. The single-layer LEO quantum satellite constellation is constructed based on the Walker-star configuration, comprising six orbital planes, each of which deploys 12 quantum satellites. All of these quantum satellites operate at an altitude of 500 km with an inclination of 88°. For the dual-layer LEO quantum satellite constellation, we combine the Walker-star and Walker-delta configurations, where the Walker-star configuration operates at an altitude of 500 km with an inclination of 88°, and the Walker-delta configuration operates at an altitude of 550 km with an inclination of 53°. To match the total number of quantum satellites of the single-layer constellation, both configurations within the dual-layer constellation comprise six orbital planes, each housing six quantum satellites. We consider a global distribution of 15 ground stations, three of which are realistic ones connected to practical LEO quantum satellites. Among these, the Beijing and Graz ground stations are selected from those connected to Micius [10], and the Stellenbosch ground station is selected from those connected to Jinan-1 [11]. The QKD device quantities n_{st}, n_{sr}, and n_{gr} are all set to 2.

Table 1. Simulation Parameters for SKR Calculation

Parameter	Value
QKD source wavelength λ	843.9 nm
Diameter of the receiver telescope D_{r}	0.6 m
Transmissivity η_{obs} of the receiver telescope due to the central obstruction	0.91
Full-angle beam divergence θ	17.2×10^{-6} rad
Atmospheric transmissivity η_λ at zenith ($\lambda = 843.9$ nm)	0.5
Beam decoherence efficiency (caused by turbulence effects) η_{turb}	0.631
Photon detection efficiency η_{pde}	0.58
Receiver optical efficiency η_{rx}	0.753
Coupling efficiency η_{cpl}	0.56
Pointing and tracking efficiency η_{pat}	0.912
Filter bandwidth $\Delta\lambda$	0.2 nm
Detection time window Δt	1 ns
Solid angle FOV Ω_{FOV}	5.383×10^{-8} sr
Overall dark count rate f_{dark}	200 Hz
Misalignment error probability	1.5%
Error correction efficiency	1.22
Repetition frequency	625 MHz

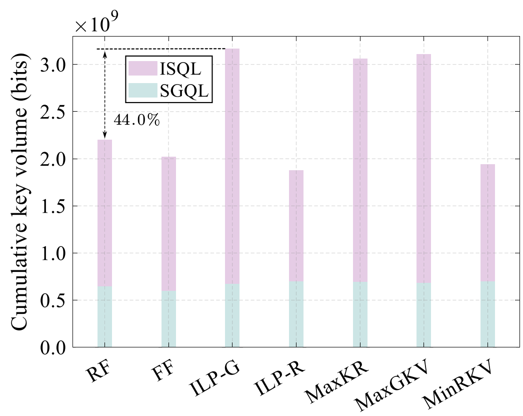
The full operation duration is set to 24 h, with an interval τ set to 1 min. This results in a total of 1441 topology snapshots, i.e., $|T| = 1441$. The simulation parameters for calculating SKRs based on the BB84 protocol are derived from [11,15,43] and listed in Table 1. Furthermore, the background radiance H_{b} varies across different illumination conditions, which is set to 10^{-4}, 10^{-3}, and 1 for the umbra, penumbra, and full solar illumination conditions [15], respectively.

We use conventional QKD link establishment schemes based on RF and FF as the benchmarks for comparison, which have been described in Subsection 3.A. In the following subsections, we compare the proposed QKD link establishment approaches against the two benchmarks in terms of the cumulative key volume, key volume distribution, quantum satellite availability, and the total number of established QKD links.

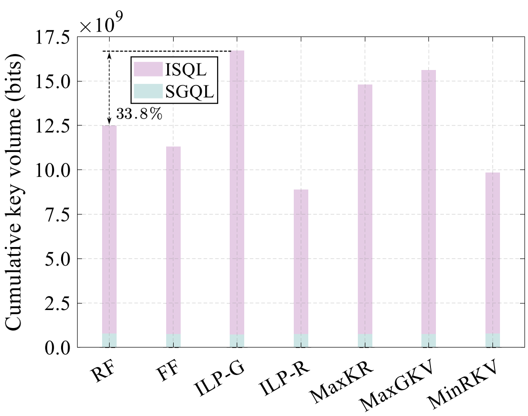
A. Cumulative Key Volume

The results of the cumulative key volume over the full operation duration for different QKD link establishment approaches, in single- and dual-layer LEO quantum satellite networks, are depicted in Figs. 6(a) and 6(b), respectively.

It can be observed that ILP-G achieves the largest cumulative key volume, followed by MaxGKV, MaxKR, RF, and FF. This is because ILP-G, MaxGKV, and MaxKR prioritize establishing QKD links with stronger key generation capabilities. Meanwhile, ILP-G fully optimizes the connection of QTxs and QRxs across the corresponding quantum satellites and ground stations. This avoids the random selection of QKD devices, thus achieving the optimal cumulative key volume. In contrast to MaxKR, MaxGKV incorporates the time-window-based key generation, which reduces reliance on fixed SKRs per time window, thereby boosting the cumulative key volume. Furthermore, the cumulative key volume of ILP-G is 44.0% and 33.8% higher than that of RF in single- and dual-layer LEO quantum satellite networks, respectively. For the two benchmarks, RF employs a random QKD link selection scheme to avoid exclusively choosing links with low serial numbers, thereby achieving more key resources than FF to some extent. Hence, the applicability of the proposed ILP models and heuristic algorithms for single- and dual-layer LEO quantum satellite networks is validated, while ILP-G, MaxGKV, and MaxKR demonstrate enhanced key generation efficiency relative to the benchmarks.



(a)



(b)

Fig. 6. Cumulative key volume over the full operation duration for different QKD link establishment approaches: (a) single-layer LEO quantum satellite network, (b) dual-layer LEO quantum satellite network.

As shown in Fig. 6, the increased key resources mainly stem from ISQLs, which underscores the importance of ISQL planning. This also leads to a considerable growth in secret keys for inter-satellite services. Meanwhile, the cumulative key volume of SGQLs exhibits only minor variations across different approaches. Even for ILP-R, which prioritizes establishing QKD links with the lowest RKV (typically SGQLs), the difference from other approaches remains limited. This indicates that these approaches do not significantly sacrifice the cumulative key volume of SGQLs by excessively favoring high-quality links, and thus do not notably affect end-to-end services between ground stations. Furthermore, the cumulative key volume in Fig. 6(b) shows a significant improvement relative to that in Fig. 6(a), especially across ISQLs. This further demonstrates that the dual-layer LEO quantum satellite constellation exhibits superior key generation capabilities and stronger collaborative relationships among quantum satellites. Notably, ILP-R and MinRKV exhibit a lower cumulative key volume than the benchmarks. Their objective is to balance key resource distribution rather than always selecting the QKD links with higher secret key production. Instead, they primarily prioritize QKD links with the smallest remaining key volume. These links typically have low key production. The advantages of ILP-R and MinRKV will be detailed in subsequent subsections.

B. Key Volume Distribution

Figures 7(a)–7(e) show the key volume distribution among nodes in single- and dual-layer LEO quantum satellite networks for different QKD link establishment approaches. These results can visually reflect the resource capacity and service provisioning capabilities. It can be observed that, for SGQLs, ground stations generate secret keys with most quantum satellites. However, the key volume distribution varies across different ground stations owing to their geographical distribution and the coverage characteristics determined by the orbital inclinations of the quantum satellites. In addition, the illumination conditions of ground stations, together with the equipment limitations and switching constraints of quantum satellites, further affect the resulting key volume distribution. For ISQLs, most key volume distributions exhibit an approximately diagonal pattern. Since the quantum satellites are numbered sequentially within each orbital plane, this pattern reflects the coordination between quantum satellites in the same orbital plane and those in other orbital planes. A continuous diagonal further indicates an approximately one-to-one correspondence between quantum satellites in one orbital plane and those in another, while more complex distribution patterns reflect greater diversity in ISQL configurations.

For the proposed QKD link establishment approaches, ILP-R and MinRKV demonstrate more balanced key volume distribution than ILP-G, MaxKR, and MaxGKV. The reason is that ILP-R and MinRKV prioritize QKD links with lower remaining key volume. This ensures as many nodes as possible can generate and accumulate key volume. Furthermore, this phenomenon is more pronounced in the single-layer LEO quantum satellite network. This verifies the superiority of ILP-R and MinRKV in balancing key resource distribution.

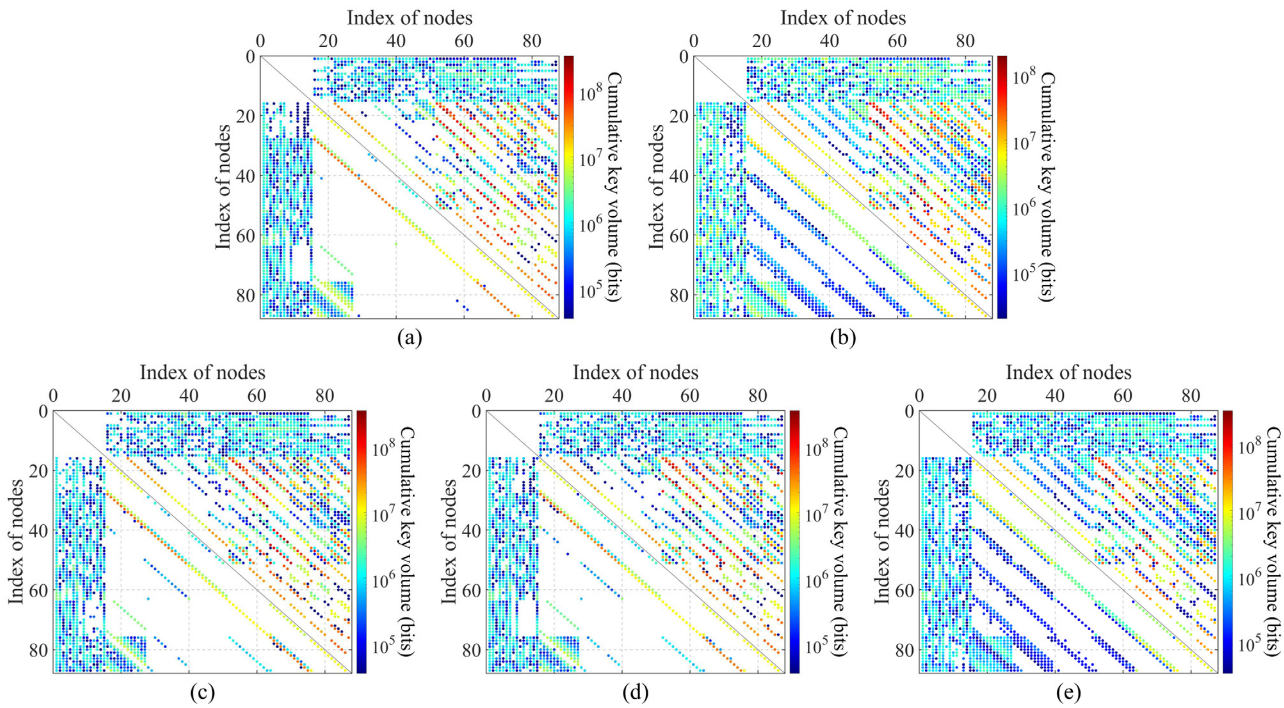


Fig. 7. Key volume distribution among nodes in single- and dual-layer LEO quantum satellite networks for different QKD link establishment approaches: (a) ILP-G, (b) ILP-R, (c) MaxKR, (d) MaxGKV, (e) MinRKV. (The upper and lower triangles represent the results of dual-layer and single-layer LEO quantum satellite networks, respectively. Node indices 1–15 represent ground stations, and indices 16–87 represent quantum satellites. White denotes node pairs with zero key volume over the full operation duration.)

It also reflects that ILP-G, MaxKR, and MaxGKV achieve enhanced key generation efficiency without considering resource balancing. This may cause negative impacts on QoS owing to insufficient key resources on some QKD links. Thus, ILP-G, MaxKR, and MaxGKV may yield lower success rates of QKD services than ILP-R and MinRKV in some practical scenarios.

In addition, the dual-layer LEO quantum satellite network exhibits more key resources generated and more balanced key resource distribution than the single-layer LEO quantum satellite network. This is because the dual-layer constellation configuration features richer satellite visibility relationships and better ground station coverage. Hence, the superiority of the dual-layer LEO quantum satellite network is demonstrated, which achieves better complementarity by combining the Walker-star and Walker-delta constellations.

C. Quantum Satellite Availability

Given that the secret keys generated by QKD links are employed for subsequent global key negotiation to meet the security requirements of multiple users, the quantum satellite availability (defined in Subsection 3.B) can reflect the coverage and interconnectivity capabilities of quantum satellites in supporting such global key negotiation processes. Figures 8 and 9 depict the quantum satellite availability via SGQLs and ISQLs, respectively, with both single- and dual-layer LEO quantum satellite networks considered. Notably, the five proposed QKD link establishment approaches and two benchmarks exhibit good applicability and scalability across different constellation configurations.

More concretely, it can be observed that ILP-R and MinRKV maintain a relatively high quantum satellite availability for most ground stations and quantum satellites. This stems from the fact that both approaches incorporate RKV considerations during QKD link establishment. They prioritize links with lower remaining key volume to enhance the network-wide quantum satellite availability. Moreover, ILP-R further excels by fully balancing the usage of QTxs and QRxs, thereby outperforming MinRKV at some nodes. For both ground stations and quantum satellites, the quantum satellite availability of RF is generally higher than that of FF at most nodes. This is because the random link selection avoids persistent preference for nodes with lower serial numbers or earlier passage. In contrast, ILP-G, MaxKR, and MaxGKV exhibit consistently lower quantum satellite availability. This stems from their persistent focus on maximizing generated key resources. Hence, they neglect relatively weaker QKD links with insufficient key resources and always select QKD links that have stronger key generation capabilities, indicating their long-term reliance on certain quantum satellites for high-volume key production.

From Figs. 8(a) and 8(b), it can be observed that the dual-layer LEO quantum satellite network exhibits a more uniform distribution of the quantum satellite availability via SGQLs. The single-layer LEO quantum satellite network, in contrast, shows a significant concentration at certain high-latitude ground stations and very low quantum satellite availability at other stations, resulting in significant distribution disparities. The reason is that the single-layer constellation, based on the

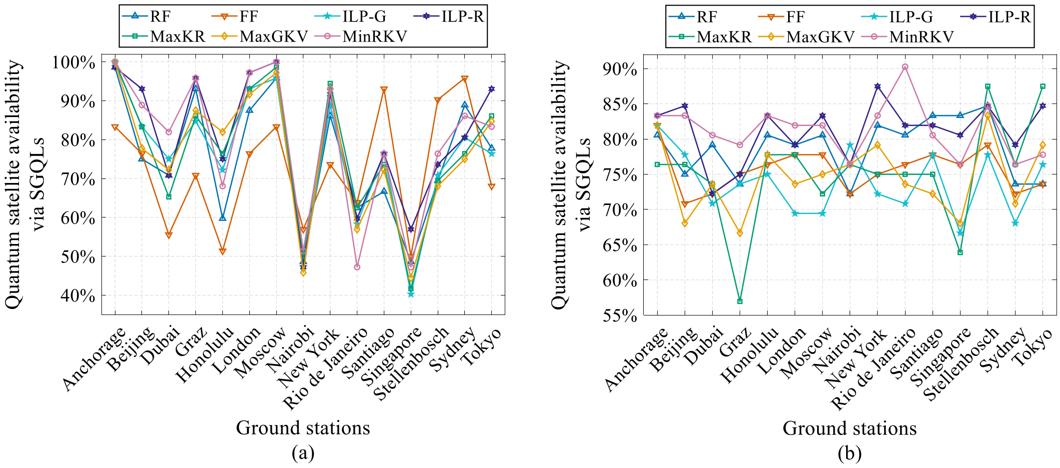


Fig. 8. Quantum satellite availability via SGQLs for different QKD link establishment approaches: (a) single-layer LEO quantum satellite network, (b) dual-layer LEO quantum satellite network.

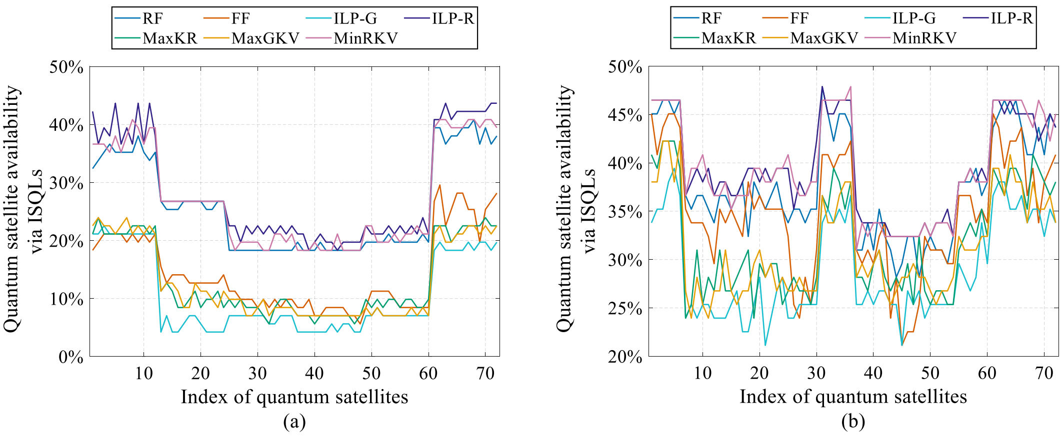


Fig. 9. Quantum satellite availability via ISQLs for different QKD link establishment approaches: (a) single-layer LEO quantum satellite network, (b) dual-layer LEO quantum satellite network.

Walker-star, features larger orbital inclinations that provide superior coverage over high-latitude regions. In contrast, the dual-layer constellation incorporates a Walker-delta constellation with a lower inclination as its second layer, thereby supplementing coverage over mid-to-low latitudes and achieving more uniform overall coverage. Similarly, the quantum satellite availability via ISQLs in Fig. 9(b) is higher than that in Fig. 9(a). This is because the dual-layer constellation configuration introduces more diverse inter-satellite connections, resulting in superior collaboration between quantum satellites compared to the single-layer constellation.

D. Total Number of Established QKD Links

The total numbers of established SGQLs and ISQLs enable the evaluation of LEO quantum satellite networks from the perspective of connection diversity and path flexibility, as illustrated in Figs. 10 and 11, respectively. For end-to-end services, maximizing only the cumulative key volume is insufficient, since the ability to support such services is also affected by the key volume available along end-to-end paths. Meanwhile, a larger total number of QKD links may provide more candidate paths, thereby improving the end-to-end service success rate. Nevertheless, the total number of QKD links alone cannot fully reflect the overall key resources, since a quantum satellite network with numerous established links may still yield a low cumulative key volume, as can be observed from Figs. 6, 10, and 11.

Similar to the quantum satellite availability analyzed in Subsection 6.C, the proposed QKD link establishment approaches and benchmarks show consistent trends. Furthermore, ILP-R and MinRKV maintain relatively more QKD links. This is because they prioritize links with low remaining key volume, which leads to the establishment of more diverse QKD links. Under the QKD device constraints, the passage of quantum satellites over ground stations exhibits periodicity for SGQLs. This results in relatively small gaps between different QKD link establishment approaches. In contrast, the establishment of ISQLs involves more complex satellite-to-satellite relationships. This enables ILP-R and MinRKV to outperform other QKD link establishment approaches. Additionally, the dual-layer LEO quantum satellite network achieves a higher average number of established

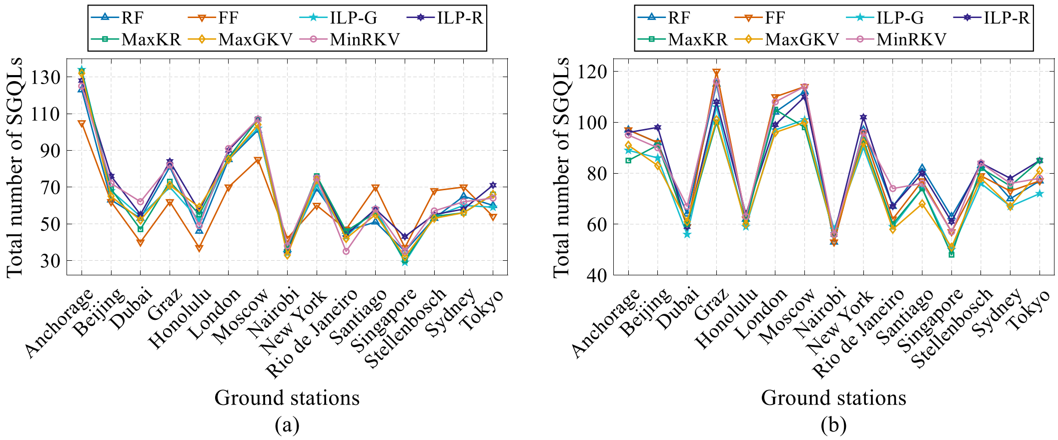


Fig. 10. Total number of established SGQLs for different QKD link establishment approaches: (a) single-layer LEO quantum satellite network, (b) dual-layer LEO quantum satellite network.

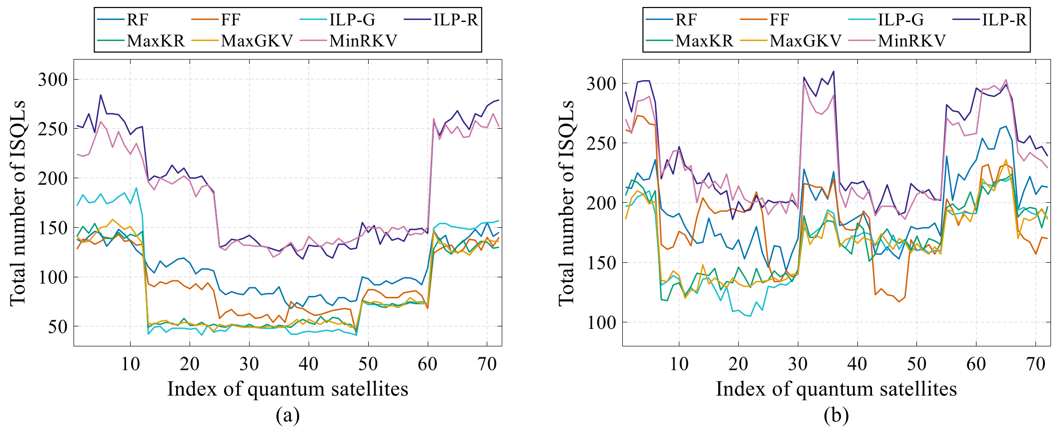


Fig. 11. Total number of established ISQLs for different QKD link establishment approaches: (a) single-layer LEO quantum satellite network, (b) dual-layer LEO quantum satellite network.

QKD links than the single-layer LEO quantum satellite network. This is because the dual-layer constellation offers more diversified satellite-to-ground and satellite-to-satellite connection options, which enable more QKD devices to be fully utilized.

Based on the above analysis, we verify that the higher cumulative key volume achieved by ILP-G and MaxGKV is not obtained through frequent link switching. In contrast, the larger total numbers of established SGQLs and ISQLs provided by ILP-R and MinRKV may improve the global balance of key resources in single- and dual-layer LEO quantum satellite networks.

7. CONCLUSION

In this paper, we address the adaptive establishment of SGQLs and ISQLs in single- and dual-layer LEO quantum satellite networks. The network model, SKR model, and SKV model are formulated, with different illumination conditions as well as diverse channel losses and receiver-side component losses incorporated. We develop ILP-G and ILP-R models to maximize the generated key volume and balance the remaining key volume for optimizing QKD link establishment, respectively, while incorporating realistic constraints. Furthermore, three adaptive QKD link establishment schemes are devised, which are implemented by the corresponding heuristic algorithms (i.e., MaxKR, MaxGKV, and MinRKV). Simulation results demonstrate the applicability and superiority of the proposed QKD link establishment approaches over two benchmarks in single- and dual-layer LEO quantum satellite networks. Among the proposed approaches, ILP-G and MaxGKV achieve the largest cumulative key volume, while ILP-R and MinRKV offer better performance in key volume distribution, quantum satellite availability, and the total number of established QKD links. In addition, the dual-layer LEO quantum satellite network exhibits more key resources generated and more balanced key resource distribution than the single-layer LEO quantum satellite network, demonstrating its better complementarity by combining the Walker-star and Walker-delta constellations. In particular, the simulation results reflect a trade-off between the enhanced key generation efficiency and the balanced key resource distribution.

In future work, maximizing GKV for dedicated services between ground stations and for inter-satellite services will be a valuable research direction. Furthermore, both insufficient and excessive link switching may degrade key generation efficiency, as infrequent link switching may lead to prolonged connections to low SKR links, while overly frequent link switching incurs high reconfiguration overheads. Such a trade-off, therefore, warrants further investigation in future research. In addition, we will discuss additional practical system aspects that could influence link establishment in real-world deployments, such as pointing acquisition time, terminal switching overhead, and satellite attitude control limitations. Meanwhile, we will explore the scalability of the proposed ILP-based optimization approach when applied to larger constellations or multi-layer architectures, as future quantum satellite networks may involve a significantly larger number of nodes.

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Data availability. Data underlying the results presented in this paper are not publicly available at this time but may be obtained from the authors upon reasonable request.

REFERENCES

1. S. S. Gill, A. Kumar, H. Singh, *et al.*, "Quantum computing: a taxonomy, systematic review and future directions," Software Pract. Exper. **52**, 66–114 (2022).
2. Z. Yang, M. Zolanvari, and R. Jain, "A survey of important issues in quantum computing and communications," IEEE Commun. Surv. Tutorials **25**, 1059–1094 (2023).
3. H.-K. Lo, M. Curty, and K. Tamaki, "Secure quantum key distribution," Nat. Photonics **8**, 595–604 (2014).
4. S. Pirandola, U. L. Andersen, L. Banchi, *et al.*, "Advances in quantum cryptography," Adv. Opt. Photonics **12**, 1012–1236 (2020).
5. Y. Cao, Y. Zhao, Q. Wang, *et al.*, "The evolution of quantum key distribution networks: on the road to the Qinternet," IEEE Commun. Surv. Tutorials **24**, 839–894 (2022).
6. C.-Y. Lu, Y. Cao, C.-Z. Peng, *et al.*, "Micius quantum experiments in space," Rev. Mod. Phys. **94**, 035001 (2022).
7. L. de F. de Parny, O. Alibart, J. Debaud, *et al.*, "Satellite-based quantum information networks: use cases, architecture, and roadmap," Commun. Phys. **6**, 12 (2023).
8. D. Orsucci, P. Kleinpaß, J. Meister, *et al.*, "Assessment of practical satellite quantum key distribution architectures for current and near-future missions," Int. J. Satell. Commun. Netw. **43**, 164–192 (2025).
9. S.-K. Liao, W.-Q. Cai, W.-Y. Liu, *et al.*, "Satellite-to-ground quantum key distribution," Nature **549**, 43–47 (2017).
10. S.-K. Liao, W.-Q. Cai, J. Handsteiner, *et al.*, "Satellite-relayed intercontinental quantum network," Phys. Rev. Lett. **120**, 030501 (2018).
11. Y. Li, W.-Q. Cai, J.-G. Ren, *et al.*, "Microsatellite-based real-time quantum key distribution," Nature **640**, 47–54 (2025).
12. L. Mazzarella, C. Lowe, D. Lowndes, *et al.*, "QUARC: quantum research CubeSat—a constellation for quantum communication," Cryptography **4**, 7 (2020).
13. T. Vergoossen, S. Loarte, R. Bedington, *et al.*, "Modelling of satellite constellations for trusted node QKD networks," Acta Astronaut. **173**, 164–171 (2020).
14. J. Wang, H. Chen, and Z. Zhu, "Modeling research of satellite-to-ground quantum key distribution constellations," Acta Astronaut. **180**, 470–481 (2021).
15. F. D. Grossi, S. Alberico, and C. Circi, "Orbit design of satellite quantum key distribution constellations in different ground stations networks," Adv. Space Res. **73**, 5446–5463 (2024).
16. Y. Li, S.-K. Liao, Y. Cao, *et al.*, "Space-ground QKD network based on a compact payload and medium-inclination orbit," Optica **9**, 933–938 (2022).
17. J. C. McDowell, "The low Earth orbit satellite population and impacts of the SpaceX Starlink constellation," Astrophys. J. Lett. **892**, L36 (2020).
18. X. Zhao, C. Wang, S. Cai, *et al.*, "Cooperative design of dual-layer LEO satellite constellation based on diversified QoS requirements and seamless multi-coverage," IEEE Trans. Veh. Technol. **74**, 925–939 (2025).
19. Z. Su, C. Liu, and X. Zhu, "A capacity analysis model for large-scale two-layered satellite networks," IEEE Trans. Commun. **73**, 8769–8780 (2025).
20. N. K. Panigrahy, P. Dhara, D. Towsley, *et al.*, "Optimal entanglement distribution using satellite based quantum networks," in *IEEE Conference on Computer Communications Workshops* (2022).
21. A. Williams, N. K. Panigrahy, A. McGregor, *et al.*, "Scalable scheduling policies for quantum satellite networks," in *IEEE International Conference on Quantum Computing and Engineering* (2024), pp.1760–1769.
22. A. Chang, Y. Wan, G. Xue, *et al.*, "Entanglement distribution in satellite-based dynamic quantum networks," IEEE Netw. **38**, 78–86 (2024).
23. X. Wei, J. Liu, L. Fan, *et al.*, "Optimal entanglement distribution problem in satellite-based quantum networks," IEEE Netw. **39**, 97–103 (2025).
24. A. Chang, Y. Wan, X. Lin, *et al.*, "Entanglement distribution in LEO satellite-based dynamic quantum networks," in *IEEE Global Communications Conference (GLOBECOM)* (2024), pp. 4485–4490.
25. R. Maule, N. K. Panigrahy, N. L. Anipeddi, *et al.*, "Fair and efficient scheduling strategies for satellite assisted quantum key distribution systems," in *IEEE International Conference on Quantum Computing and Engineering* (2024), pp. 1788–1798.
26. C. H. Bennett and G. Brassard, "Quantum cryptography: public key distribution and coin tossing," in *IEEE International Conference on Computers Systems and Signal Processing* (1984), pp. 175–179.
27. V. Karavias, C. White, A. Lord, *et al.*, "Optimizing satellite and core networks for a global quantum network," J. Opt. Commun. Netw. **16**, 504–515 (2024).
28. R. Yehia, M. Schiavon, V. M. Acosta, *et al.*, "Connecting quantum cities: simulation of a satellite-based quantum network," New J. Phys. **26**, 073015 (2024).
29. M. Guo, Y. Cao, J. Zhu, *et al.*, "Topology abstraction-based routing scheme for secret-key provisioning in hybrid GEO/LEO quantum satellite networks," Entropy **25**, 1047 (2023).
30. X. He, L. Li, D. Han, *et al.*, "Routing and secret key assignment for secure multicast services in quantum satellite networks," J. Opt. Commun. Netw. **14**, 190–203 (2022).
31. Y.-A. Chen, Q. Zhang, T.-Y. Chen, *et al.*, "An integrated space-to-ground quantum communication network over 4,600 kilometres," Nature **589**, 214–219 (2021).
32. S.-K. Liao, H.-L. Yong, C. Liu, *et al.*, "Long-distance free-space quantum key distribution in daylight towards inter-satellite communication," Nat. Photonics **11**, 509–513 (2017).
33. Z. C. M. Davidson, E. Hugues-Salas, G. M. Bonner, *et al.*, "AirQKD: the role of free-space optics quantum key distribution enabling pragmatic secure and scalable communications," IEEE Commun. Mag. **62**(10), 40–47 (2024).
34. W.-Q. Cai, Y. Li, B. Li, *et al.*, "Free-space quantum key distribution during daylight and at night," Optica **11**, 647–652 (2024).
35. P. Zhang, R. D. Oliveira, Z. C. M. Davidson, *et al.*, "Daylight quantum key distribution over free-space optics for future security networks," J. Opt. Commun. Netw. **17**, B61–B70 (2025).
36. M. Avesani, L. Calderaro, M. Schiavon, *et al.*, "Full daylight quantum-key-distribution at 1550 nm enabled by integrated silicon photonics," npj Quantum Inf. **7**, 93 (2021).
37. A. K. Ekert, "Quantum cryptography based on Bell's theorem," Phys. Rev. Lett. **67**, 661–663 (1991).
38. Y. Su, Y. Liu, Y. Zhou, *et al.*, "Broadband LEO satellite communications: architectures and key technologies," IEEE Wirel. Commun. **26**, 55–61 (2019).
39. M. Sun, T. Zhang, and L. Liu, "Constellation configuration optimization method for user distribution characteristics of LEO satellite networks," IEEE Trans. Netw. Serv. Manag. **22**, 1232–1246 (2025).

40. D. Gottesman, H.-K. Lo, N. Lütkenhaus, *et al.*, “Security of quantum key distribution with imperfect devices,” Quantum Inf. Comput. **4**, 325–360 (2004).
41. X. Ma, B. Qi, Y. Zhao, *et al.*, “Practical decoy state for quantum key distribution,” Phys. Rev. A **72**, 012326 (2005).
42. H.-K. Lo, X. Ma, and K. Chen, “Decoy state quantum key distribution,” Phys. Rev. Lett. **94**, 230504 (2005).
43. T. Roger, R. Singh, C. Perumangatt, *et al.*, “Real-time gigahertz free-space quantum key distribution within an emulated satellite overpass,” Sci. Adv. **9**, eadj5873 (2023).
44. G. Wang, F. Yang, J. Song, *et al.*, “Free space optical communication for inter-satellite link: architecture, potentials and trends,” IEEE Commun. Mag. **62**(3), 110–116 (2024).
45. M. T. Gruneisen, M. B. Flanagan, B. A. Sickmiller, *et al.*, “Modeling daytime sky access for a satellite quantum key distribution downlink,” Opt. Express **23**, 23924–23934 (2015).
46. N. Razmi, B. Matthiesen, A. Dekorsy, *et al.*, “On-board federated learning for satellite clusters with inter-satellite links,” IEEE Trans. Commun. **72**, 3408–3424 (2024).
47. T. Xiong, R. Zhang, C. Luo, *et al.*, “A novel scheduling scheme for earth observation in LEO satellite systems,” in *IEEE Global Communications Conference (GLOBECOM)* (2023), pp. 2305–2310.